

1 Assignment Overview

What you will implement

1. Byte-pair encoding (BPE) tokenizer
2. Transformer language model (LM)
3. The cross-entropy loss function and the AdamW optimizer
4. The training loop, with support for serializing and loading model and optimizer state

DONE

What you will run

1. Train a BPE tokenizer on the TinyStories dataset.
2. Run your trained tokenizer on the dataset to convert it into a sequence of integer IDs.
3. Train a Transformer LM on the TinyStories dataset.
4. Generate samples and evaluate perplexity using the trained Transformer LM.
5. Train models on OpenWebText and submit your attained perplexities to a leaderboard.

DONE

2 Byte-Pair Encoding (BPE) Tokenizer

2.1 The Unicode Standard

In Python, you can use the `ord()` function to convert a single Unicode character into its integer representation. The `chr()` function converts an integer Unicode code point into a string with the corresponding character.

Problem (unicode1): Understanding Unicode (1 point)

- (a) What Unicode character does `chr(0)` return?

Answer: `'\x00'`

- (b) How does this character's string representation `__repr__()` differ from its printed representation?

Answer: The `__repr__()` is designed to be **unambiguous** and is mainly for developers, showing the technical details of the object. On the other hand, the printed representation is meant to be **readable** and user-friendly.

- (c) What happens when this character occurs in text? It may be helpful to play around with the following in your Python interpreter and see if it matches your expectations:

Answer: Code below.

```
>>> chr(0)
'\x00'
>>> print(chr(0))

>>> "this is a test" + chr(0) + "string"
'this is a test\x00string'
>>> print("this is a test" + chr(0) + "string")
this is a teststring
```

2.2 Unicode Encodings

- UTF-8: The Efficient Variable-Length Encoder, It uses 1 to 4 bytes per character.
- UTF-16: The Middle Ground, It uses either 2 or 4 bytes per character.
- UTF-32: The Simple Fixed-Length Encoder, It uses exactly 4 bytes for every character, regardless of what it is.

Problem (unicode2): Unicode Encodings (3 point)

- (a) What are some reasons to prefer training our tokenizer on UTF-8 encoded bytes, rather than UTF-16 or UTF-32? It may be helpful to compare the output of these encodings for various input strings.

Answer: `'\x00'`

- (b) Consider the following (incorrect) function, which is intended to decode a UTF-8 byte string into a Unicode string. Why is this function incorrect? Provide an example of an input byte string that yields incorrect results.

```
def decode_utf8_bytes_to_str_wrong(bytestring: bytes):  
    return "".join([bytes([b]).decode("utf-8") for b in bytestring])  
  
>>> decode_utf8_bytes_to_str_wrong("hello".encode("utf-8"))  
'hello'
```

Example: `b'\xe4\xb8\xad'`.

Answer: This function is incorrect because it attempts to decode each byte individually, which fails for any character encoded as a multi-byte sequence, resulting in a `UnicodeDecodeError`.

- (c) Give a two byte sequence that does not decode to any Unicode character(s).

Example: `b'\xff\xff'`.

Answer: This sequence is invalid because the byte `0xFF` is explicitly forbidden in the UTF-8 standard and does not correspond to any valid lead or continuation byte in the encoding scheme.

2.3 Subword Tokenization

Sennrich et al. [2016] propose to use byte-pair encoding (BPE; Gage, 1994), a compression algorithm that iteratively replaces (“merges”) the most frequent pair of bytes with a single, new unused index.

2.4 BPE Tokenizer Training

Vocabulary initialization

Pre-tokenization Use regular expressions to split the sentence into independent punctuation marks and independent items that may include spaces, and improve the merging effect by counting the occurrence frequency of these independent items.

```
>>> PAT = r"'(?:[sdmt]|ll|ve|re)| ?\p{L}+| ?\p{N}+| ?[\^\s\p{L}\p{N}]+\s+(?!S)|\s+"
>>> # requires `regex` package
>>> import regex as re
>>> re.findall(PAT, "some text that i'll pre-tokenize")
['some', ' text', ' that', ' i', "'ll", ' pre', '-', 'tokenize']
```

Compute BPE merges During the BPE merging process, two **principles** must be followed:

- Do not **cross** pre-tokenization boundaries (this is already guaranteed algorithmically);
- Handle issues with a **reproducible conflict resolution strategy**, such as selecting the lexicographically largest word pair to merge when the occurrence counts are the same.

Special tokens

2.5 Experimenting with BPE Tokenizer Training

Parallelizing pre-tokenization Using the provided segmentation code and the `multiprocessing` library, parallelize the tokenization statistics.

Removing special tokens before pre-tokenization

Optimizing the merging step After performing a statistical pass to obtain a tokenization table with frequencies, perform merging only within that tokenization table.

Low-Resource/Downscaling Tip: Profiling

You should use profiling tools like `cProfile` or `scalene` to identify the bottlenecks in your implementation, and focus on optimizing those.

Problem (train_bpe): BPE Tokenizer Training (15 points)

Deliverable: Write a function that, given a path to an input text file, trains a (byte-level) BPE tokenizer. Your BPE training function should handle (at least) the following input parameters:

input_path: `str` Path to a text file with BPE tokenizer training data.

vocab_size: `int` A positive integer that defines the maximum final vocabulary size (including the initial byte vocabulary, vocabulary items produced from merging, and any special tokens).

special_tokens: `list[str]` A list of strings to add to the vocabulary. These special tokens do not otherwise affect BPE training.

Your BPE training function should return the resulting vocabulary and merges:

vocab: `dict[int, bytes]` The tokenizer vocabulary, a mapping from `int` (token ID in the vocabulary) to `bytes` (token bytes).

merges: `list[tuple[bytes, bytes]]` A list of BPE merges produced from training. Each list item is a `tuple` of `bytes` (`<token1>`, `<token2>`), representing that `<token1>` was merged with `<token2>`. The merges should be ordered by order of creation.

Insights: BPE Tokenizer Training

Python plus Rust: Python is used for rapid prototyping, while Rust handles the actual deployment of the product.

Glitch Tokens: Due to insufficiently strict data cleaning, some meaningless words—namely **Mojibake**—may appear during the BPE training process. Moreover, because of BPE’s **greedy, unsupervised, and non-semantic nature**, a large number of such meaningless word merges occur, turning them into abnormally long and nonsensical words.

Vocabulary Bloat (Dead Tokens): Because BPE is incremental and never removes merges, the intermediate products of long words permanently occupy the vocabulary, leading to wasted model parameters and tokenization fragmentation. In industry, there are three approaches to address this issue: First, absolutely aggressive data cleaning; second, stricter regex filtering. Both of these are common methods for limiting glitch tokens. The third approach is to switch to the **Unigram** algorithm, which is frequently used by Google-family models (such as T5 and some Llama variants).

Unigram: Unigram works by "subtraction": it first initializes an extremely large vocabulary (e.g., 100,000 words) and **then calculates how much the overall corpus compression rate would be affected if a given word were removed**. Those useless intermediate composite words are directly eliminated and discarded during training.

Problem (train_bpe_tinystories): BPE Training on TinyStories (2 points)

- (a) Train a byte-level BPE tokenizer on the TinyStories dataset, using a maximum vocabulary size of 10,000. Make sure to add the TinyStories `<|endoftext|>` special token to the vocabulary. Serialize the resulting vocabulary and merges to disk for further inspection. How many hours and memory did training take? What is the longest token in the vocabulary? Does it make sense?

Resource requirements: ≤ 30 minutes (no GPUs), ≤ 30 GB RAM

Resource used: 10.78 seconds, 2.6GB RAM

Longest token: 'disappointment'

- (b) Profile your code. What part of the tokenizer training process takes the most time?

Answer: `process_chunk_with_special_tokens` takes up the vast majority of the time.

Problem (train_bpe_expts_owt): BPE Training on OpenWebText (2 points)

- (a) Train a byte-level BPE tokenizer on the OpenWebText dataset, using a maximum vocabulary size of 32,000. Serialize the resulting vocabulary and merges to disk for further inspection. What is the longest token in the vocabulary? Does it make sense?

Resource requirements: ≤ 12 hours (no GPUs), ≤ 100 GB RAM

Resource used: 6 minutes, 7.0GB RAM

Longest token: '_____'

- (b) Compare and contrast the tokenizer that you get training on TinyStories versus OpenWebText.

Answer: Because TinyStories has an extremely limited vocabulary, a 10k vocab size forces the BPE tokenizer to over-merge, creating tokens that represent entire multi-word phrases. In contrast, OpenWebText's diverse internet text requires the BPE algorithm to rely heavily on subwords (like prefixes and suffixes) to represent its massive vocabulary within a 32k limit, resulting in shorter, more fragmented tokens for rare words.

2.6 BPE Tokenizer: Encoding and Decoding

2.6.1 Encoding text

Special tokens.

Memory considerations.

2.6.2 Decoding text

Problem (tokenizer): Implementing the tokenizer (15 points)

Deliverable: Implement a `Tokenizer` class that, given a vocabulary and a list of merges, encodes text into integer IDs and decodes integer IDs into text. Your tokenizer should also support user-provided special tokens (appending them to the vocabulary if they aren't already there).

Answer: code in appendix.

2.7 Experiments

Problem (tokenizer_experiments): Experiments with tokenizers (4 points)

- (a) Sample 10 documents from TinyStories and OpenWebText. Using your previously-trained TinyStories and OpenWebText tokenizers (10K and 32K vocabulary size, respectively), encode these sampled documents into integer IDs. What is each tokenizer's compression ratio (bytes/token)?

Answer: The TinyStories tokenizer achieves a higher compression ratio (5.5 bytes/token) because its vocabulary easily memorizes long, frequent whole words from its restricted domain. In contrast, the OpenWebText tokenizer has a lower compression ratio (3.8 bytes/token) as it must rely heavily on shorter subwords to encode a highly diverse vocabulary.

- (b) What happens if you tokenize your OpenWebText sample with the TinyStories tokenizer? Compare the compression ratio and/or qualitatively describe what happens.

Answer: Tokenizing OpenWebText with the TinyStories tokenizer causes a severe domain mismatch, forcing the tokenizer to aggressively fragment unfamiliar web text into individual characters or very small byte sequences. Consequently, the token count explodes, resulting in a dramatically lower (worse) compression ratio compared to using the native OpenWebText tokenizer.

- (c) Estimate the throughput of your tokenizer (e.g., in bytes/second). How long would it take to tokenize the Pile dataset (825GB of text)?

Answer: The throughput of the tokenizer depends on the implementation details, but a rough estimate for a modern CPU might be around 100 MB/second. Given the Pile dataset's size of 825GB, it would take approximately 8.25 hours to tokenize the entire dataset.

- (d) Using your TinyStories and OpenWebText tokenizers, encode the respective training and development datasets into a sequence of integer token IDs. We'll use this later to train our language model. We recommend serializing the token IDs as a NumPy array of datatype `uint16`. Why is `uint16` an appropriate choice?

Answer: `uint16` is an appropriate choice because it can represent all possible token IDs within the range of 0 to 65,535, which is sufficient for most practical applications and allows for efficient memory usage.

3 Transformer Language Model Architecture

3.1 Transformer LM

3.1.1 Token Embeddings

3.1.2 Pre-norm Transformer Block

3.2 Output Normalization and Embedding

3.3 Remark: Batching, Einsum and Efficient Computation

3.3.1 Mathematical Notation and Memory Ordering

3.4 Basic Building Blocks: Linear and Embedding Modules

3.4.1 Parameter Initialization

3.4.2 Linear Module

Problem (linear): Implementing the linear module

(1 point)

Deliverable: Implement a `Linear` class that inherits from `torch.nn.Module` and performs a linear transformation. Your implementation should follow the interface of PyTorch's built-in `nn.Linear` module, except for not having a `bias` argument or parameter. We recommend the following interface:

`def __init__(self, in_features, out_features, device=None, dtype=None)` Construct a linear transformation module. This function should accept the following parameters:

`in_features: int` final dimension of the input

`out_features: int` final dimension of the output

`device: torch.device | None = None` Device to store the parameters on

`dtype: torch.dtype | None = None` Data type of the parameters

`def forward(self, x: torch.Tensor) -> torch.Tensor` Apply the linear transformation to the input.

Make sure to:

- subclass `nn.Module`
- call the superclass constructor
- construct and store your parameter as W (not W^T) for memory ordering reasons, putting it in an `nn.Parameter`
- of course, don't use `nn.Linear` or `nn.functional.linear`

Code below;

3.4.3 Embedding Module

Problem (embedding): Implement the embedding module**(1 point)**

Deliverable: Implement the `Embedding` class that inherits from `torch.nn.Module` and performs an embedding lookup. Your implementation should follow the interface of PyTorch's built-in `nn.Embedding` module. We recommend the following interface:

```
def __init__(self, num_embeddings, embedding_dim, device=None, dtype=None)
```

Construct an embedding module. This function should accept the following parameters:

`num_embeddings`: `int` Size of the vocabulary

`embedding_dim`: `int` Dimension of the embedding vectors, i.e., d_{model}

`device`: `torch.device` | `None` = `None` Device to store the parameters on

`dtype`: `torch.dtype` | `None` = `None` Data type of the parameters

```
def forward(self, token_ids: torch.Tensor) -> torch.Tensor
```

Lookup the embedding vectors for the given token IDs.

Make sure to:

- subclass `nn.Module`
- call the superclass constructor
- initialize your embedding matrix as a `nn.Parameter`
- store the embedding matrix with the `d_model` being the final dimension
- of course, don't use `nn.Embedding` or `nn.functional.embedding`

Again, use the settings from above for initialization, and use `torch.nn.init.trunc_normal_` to initialize the weights.

To test your implementation, implement the test adapter at `[adapters.run_embedding]`. Then, run `uv run pytest -k test_embedding`.

3.5 Pre-Norm Transformer Block

3.5.1 Root Mean Square Layer Normalization

Problem (rmsnorm): Root Mean Square Layer Normalization (1 point)

Deliverable: Implement RMSNorm as a `torch.nn.Module`. We recommend the following interface:

```
def __init__(self, d_model: int, eps: float = 1e-5, device=None, dtype=None)
```

Construct the RMSNorm module. This function should accept the following parameters:

`d_model: int` Hidden dimension of the model

`eps: float = 1e-5` Epsilon value for numerical stability

`device: torch.device | None = None` Device to store the parameters on

`dtype: torch.dtype | None = None` Data type of the parameters

```
def forward(self, x: torch.Tensor) -> torch.Tensor
```

Process an input tensor of shape `(batch_size, sequence_length, d_model)` and return a tensor of the same shape.

Note: Remember to upcast your input to `torch.float32` before performing the normalization (and later downcast to the original dtype), as described above.

To test your implementation, implement the test adapter at `[adapters.run_rmsnorm]`. Then, run `uv run pytest -k test_rmsnorm`.

3.5.2 Position-Wise Feed-Forward Network

Problem (positionwise_feedforward): Implement the position-wise feed-forward network (2 points)

Deliverable: Implement the SwiGLU feed-forward network, composed of a SiLU activation function and a GLU.

Note: in this particular case, you should feel free to use `torch.sigmoid` in your implementation for numerical stability.

You should set d_{ff} to approximately $\frac{8}{3} \times d_{\text{model}}$ in your implementation, while ensuring that the dimensionality of the inner feed-forward layer is a multiple of 64 to make good use of your hardware. To test your implementation against our provided tests, you will need to implement the test adapter at `[adapters.run_swiglu]`. Then, run `uv run pytest -k test_swiglu` to test your implementation.

3.5.3 Relative Positional Embeddings

Problem (rope): Implement RoPE (2 points)

Deliverable: Implement a class `RotaryPositionalEmbedding` that applies RoPE to the input tensor. The following interface is recommended:

```
def __init__(self, theta: float, d_k: int, max_seq_len: int, device=None)
```

Construct the RoPE module and create buffers if needed.

`theta: float` Θ value for the RoPE

`d_k: int` dimension of query and key vectors

`max_seq_len: int` Maximum sequence length that will be inputted

`device: torch.device | None = None` Device to store the buffer on

```
def forward(self, x: torch.Tensor, token_positions: torch.Tensor) -> torch.Tensor
```

Process an input tensor of shape $(\dots, \text{seq_len}, d_k)$ and return a tensor of the same shape. Note that you should tolerate x with an arbitrary number of batch dimensions. You should assume that the token positions are a tensor of shape $(\dots, \text{seq_len})$ specifying the token positions of x along the sequence dimension.

You should use the token positions to slice your (possibly precomputed) `cos` and `sin` tensors along the sequence dimension.

To test your implementation, complete `[adapters.run_rope]` and make sure it passes `uv run pytest -k test_rope`.

3.5.4 Scaled Dot-Product Attention

Problem (softmax): Implement softmax

(1 point)

Deliverable: Write a function to apply the softmax operation on a tensor. Your function should take two parameters: a tensor and a *dimension* i , and apply softmax to the i -th dimension of the input tensor. The output tensor should have the same shape as the input tensor, but its i -th dimension will now have a normalized probability distribution. Use the trick of subtracting the maximum value in the i -th dimension from all elements of the i -th dimension to avoid numerical stability issues.

To test your implementation, complete `[adapters.run_softmax]` and make sure it passes `uv run pytest -k test_softmax_matches_pytorch`.

Problem (scaled_dot_product_attention): Implement scaled dot-product attention (5 points)

Deliverable: Implement the scaled dot-product attention function. Your implementation should handle keys and queries of shape `(batch_size, ..., seq_len, d_k)` and values of shape `(batch_size, ..., seq_len, d_v)`, where `...` represents any number of other batch-like dimensions (if provided). The implementation should return an output with the shape `(batch_size, ..., d_v)`. See section 3.3 for a discussion on batch-like dimensions.

Your implementation should also support an optional user-provided boolean mask of shape `(seq_len, seq_len)`. The attention probabilities of positions with a mask value of `True` should collectively sum to 1, and the attention probabilities of positions with a mask value of `False` should be zero.

To test your implementation against our provided tests, you will need to implement the test adapter at `[adapters.run_scaled_dot_product_attention]`.

`uv run pytest -k test_scaled_dot_product_attention` tests your implementation on third-order input tensors, while `uv run pytest -k test_4d_scaled_dot_product_attention` tests your implementation on fourth-order input tensors.

3.5.5 Causal Multi-Head Self-Attention

A Complete Source Code

A.1 Fast BPE Implementation (Rust)

```

1  use pyo3::prelude::*;
2  use std::collections::{HashMap, HashSet, BinaryHeap};
3  use std::cmp::Ordering;
4
5  #[derive(Eq, PartialEq)]
6  struct HeapItem {
7      count: usize,
8      pair: (u32, u32),
9  }
10
11  impl Ord for HeapItem {
12      fn cmp(&self, other: &Self) -> Ordering {
13          self.count.cmp(&other.count)
14              .then_with(|| self.pair.cmp(&other.pair))
15      }
16  }
17
18  impl PartialOrd for HeapItem {
19      fn partial_cmp(&self, other: &Self) -> Option<Ordering> {
20          Some(self.cmp(other))
21      }
22  }
23
24  #[pyfunction]
25  fn train_bpe_rust(
26      global_counts: HashMap<Vec<u8>, usize>,
27      target_vocab_size: usize,
28  ) -> PyResult<(Vec<(Vec<u8>, Vec<u8>>), HashMap<usize, Vec<u8>>>) {
29
30      let mut vocab: HashMap<u32, Vec<u8>> = (0..256)
31          .map(|i| (i as u32, vec![i as u8]))
32          .collect();
33      let mut next_id = 256u32;
34      let num_merges = target_vocab_size.saturating_sub(256);
35      let mut merges = Vec::with_capacity(num_merges);
36
37      let mut words: Vec<Vec<u32>> = Vec::new();
38      let mut word_freqs: Vec<usize> = Vec::new();
39
40      for (bytes, freq) in global_counts {
41          words.push(bytes.into_iter().map(|b| b as u32).collect());
42          word_freqs.push(freq);
43      }
44
45      let mut pair_counts: HashMap<(u32, u32), usize> = HashMap::new();
46      let mut pair_to_words: HashMap<(u32, u32), HashSet<usize>> = HashMap::new();
47
48      for (idx, word_ids) in words.iter().enumerate() {
49          let freq = word_freqs[idx];
50          for pair in word_ids.windows(2) {
51              let p = (pair[0], pair[1]);
52              *pair_counts.entry(p).or_insert(0) += freq;
53              pair_to_words.entry(p).or_insert_with(HashSet::new).insert(idx);
54          }

```

```

55     }
56
57     let mut heap: BinaryHeap<HeapItem> = pair_counts.iter()
58         .map(|(&pair, &count)| HeapItem { count, pair })
59         .collect();
60
61     for _ in 0..num_merges {
62         let mut best_pair = None;
63         while let Some(top) = heap.pop() {
64             if let Some(&current_count) = pair_counts.get(&top.pair) {
65                 if current_count == top.count {
66                     best_pair = Some(top.pair);
67                     break;
68                 }
69             }
70         }
71
72         let p = match best_pair {
73             Some(p) => p,
74             None => break,
75         };
76
77         let p1_bytes = vocab.get(&p.0).unwrap().clone();
78         let p2_bytes = vocab.get(&p.1).unwrap().clone();
79         let mut new_token_bytes = p1_bytes;
80         new_token_bytes.extend(p2_bytes);
81
82         merges.push((vocab.get(&p.0).unwrap().clone(), vocab.get(&p.1).unwrap().clone()));
83         vocab.insert(next_id, new_token_bytes);
84
85         let new_id = next_id;
86         next_id += 1;
87
88         if let Some(word_indices) = pair_to_words.remove(&p) {
89             pair_counts.remove(&p);
90
91             for &word_idx in &word_indices {
92                 let freq = word_freqs[word_idx];
93                 let old_ids = &words[word_idx];
94                 let mut new_ids = Vec::new();
95
96                 let mut i = 0;
97                 while i < old_ids.len() {
98                     if i < old_ids.len() - 1 && old_ids[i] == p.0 && old_ids[i+1] == p.1 {
99                         new_ids.push(new_id);
100                         i += 2;
101                     } else {
102                         new_ids.push(old_ids[i]);
103                         i += 1;
104                     }
105                 }
106
107                 for pair in old_ids.windows(2) {
108                     let old_p = (pair[0], pair[1]);
109                     if let Some(count) = pair_counts.get_mut(&old_p) {
110                         *count -= freq;
111                     }
112                 }
113             }

```

```

114         words[word_idx] = new_ids;
115         let updated_ids = &words[word_idx];
116         for pair in updated_ids.windows(2) {
117             let new_p = (pair[0], pair[1]);
118             let count = pair_counts.entry(new_p).or_insert(0);
119             *count += freq;
120             pair_to_words.entry(new_p).or_insert_with(HashSet::new).insert(word_idx);
121             heap.push(HeapItem { count: *count, pair: new_p });
122         }
123     }
124 }
125 }
126
127 let final_vocab = vocab.into_iter().map(|(k, v)| (k as usize, v)).collect();
128 Ok((merges, final_vocab))
129 }
130
131 #[pymodule]
132 fn fast_bpe(m: &Bound<'_, pyo3::types::PyModule>) -> PyResult<()> {
133     m.add_function(wrap_pyfunction!(train_bpe_rust, m)?);
134     Ok(())
135 }

```

A.2 Python BPE Trainer

```

1  import os
2  import time
3  import pickle
4  import regex as re
5  import multiprocessing
6  from typing import BinaryIO
7  from collections import Counter
8
9  import fast_bpe
10
11  # =====
12  # 1. 全局配置与正则定义
13  # =====
14
15  PATTERN = r"('?:[sdmt]|ll|ve|re)| ?\p{L}+| ?\p{N}+| ?[\s\p{L}\p{N}]+\s+(?!S)\s+"
16  PRETOKEN_REGEX = re.compile(PATTERN)
17
18  # =====
19  # 2. 多进程预分词辅助函数 (保持原样, Python 的强项)
20  # =====
21
22  def process_chunk_with_special_tokens(chunk_text: str, special_tokens: list[str], pretoken_regex: re.Pattern) ->
↳ Counter:
23      split_pattern = "|".join(map(re.escape, special_tokens)) if special_tokens else "(?!)"
24      text_segments = re.split(split_pattern, chunk_text)
25      chunk_counts = Counter()
26      for segment in text_segments:
27          if not segment: continue
28          for match in pretoken_regex.finditer(segment):
29              pre_token = match.group()
30              byte_token = pre_token.encode("utf-8")
31              chunk_counts[byte_token] += 1
32      return chunk_counts
33
34  def find_chunk_boundaries(file: BinaryIO, desired_num_chunks: int, split_special_token: bytes) -> list[int]:
35      if not split_special_token:
36          file.seek(0, os.SEEK_END)
37          file_size = file.tell()
38          chunk_size = file_size // desired_num_chunks
39          boundaries = [i * chunk_size for i in range(desired_num_chunks + 1)]
40          boundaries[-1] = file_size
41          return sorted(set(boundaries))
42
43      assert isinstance(split_special_token, bytes)
44      file.seek(0, os.SEEK_END)
45      file_size = file.tell()
46      file.seek(0)
47      chunk_size = file_size // desired_num_chunks
48      chunk_boundaries = [i * chunk_size for i in range(desired_num_chunks + 1)]
49      chunk_boundaries[-1] = file_size
50      mini_chunk_size = 4096
51      for bi in range(1, len(chunk_boundaries) - 1):
52          initial_position = chunk_boundaries[bi]
53          file.seek(initial_position)
54          while True:
55              mini_chunk = file.read(mini_chunk_size)
56              if mini_chunk == b"":

```

```

57         chunk_boundaries[bi] = file_size
58         break
59     found_at = mini_chunk.find(split_special_token)
60     if found_at != -1:
61         chunk_boundaries[bi] = initial_position + found_at
62         break
63     initial_position += mini_chunk_size
64     return sorted(set(chunk_boundaries))
65
66 def worker_task(file_path: str, start: int, end: int, special_tokens: list[str]) -> Counter:
67     with open(file_path, "rb") as f:
68         f.seek(start)
69         chunk_bytes = f.read(end - start)
70         chunk_text = chunk_bytes.decode("utf-8", errors="ignore")
71         return process_chunk_with_special_tokens(chunk_text, special_tokens, PRETOKEN_REGEX)
72
73 def run_parallel_pretokenization(input_path: str, special_tokens: list[str]) -> Counter:
74     num_workers = multiprocessing.cpu_count()
75     num_tasks = num_workers * 4
76     boundary_token = special_tokens[0].encode('utf-8') if special_tokens else b""
77     with open(input_path, "rb") as f:
78         boundaries = find_chunk_boundaries(f, num_tasks, boundary_token)
79     with multiprocessing.Pool(processes=num_workers) as pool:
80         tasks = [(input_path, start, end, special_tokens) for start, end in zip(boundaries[:-1], boundaries[1:])]
81         partial_counts = pool.starmap(worker_task, tasks)
82     global_counts = Counter()
83     for local_count in partial_counts:
84         global_counts.update(local_count)
85     return global_counts
86
87
88 # =====
89 # 3. 核心大作业交付函数: train_bpe
90 # =====
91
92 def train_bpe(input_path: str, vocab_size: int, special_tokens: list[str]) -> tuple[dict[int, bytes],
93 ↪ list[tuple[bytes, bytes]]]:
94     print(" 开始预分词与频率统计 (Python 多进程)...")
95     global_counts = run_parallel_pretokenization(input_path, special_tokens)
96
97     num_merges = vocab_size - 256 - len(special_tokens)
98     if num_merges < 0:
99         raise ValueError("vocab_size is too small to fit base bytes and special tokens.")
100
101     print(f" 将数据移交 Rust 底层进行合并 (目标总词表大小: {vocab_size - len(special_tokens)})...")
102
103     merges, vocab = fast_bpe.train_bpe_rust(global_counts, vocab_size - len(special_tokens))
104
105     print("Rust 计算完成, 添加 Special Tokens...")
106
107     vocab_idx = 256 + len(merges)
108     for st in special_tokens:
109         vocab[vocab_idx] = st.encode("utf-8")
110         vocab_idx += 1
111
112     print("BPE 训练彻底完成!")
113     return vocab, merges
114

```

```

115 # =====
116 # 测试/执行入口
117 # =====
118 if __name__ == "__main__":
119
120     file_path = "/home/inferior/Projects/CS336/data/owt_train.txt"
121     # file_path = "/home/inferior/Projects/CS336/data/TinyStoriesV2-GPT4-train.txt"
122     target_vocab_size = 32000
123     # target_vocab_size = 10000
124     special_tks = ["<|endoftext|>"]
125
126     print("--- 开始 BPE 训练任务 ---")
127     start_time = time.time()
128
129     vocab, merges = train_bpe(
130         input_path=file_path,
131         vocab_size=target_vocab_size,
132         special_tokens=special_tks
133     )
134
135     end_time = time.time()
136     print(f"\n 训练总耗时: {end_time - start_time:.2f} 秒")
137     print(f" 最终合并列表长度: {len(merges)}")
138     print(f" 最终词表大小: {len(vocab)}")
139
140     # -----
141     # 序列化保存 (Serialization)
142     # -----
143     output_dir = "tokenizer_outputs"
144     os.makedirs(output_dir, exist_ok=True)
145
146     vocab_path = os.path.join(output_dir, "vocab.pkl")
147     merges_path = os.path.join(output_dir, "merges.pkl")
148
149     with open(vocab_path, "wb") as f:
150         pickle.dump(vocab, f)
151
152     with open(merges_path, "wb") as f:
153         pickle.dump(merges, f)
154
155     print(f" 序列化完成! 文件已保存至: {output_dir}/")
156
157     # -----
158     # 分析: 寻找最长的 Token (作业要求)
159     # -----
160     longest_token_id = max(vocab, key=lambda k: len(vocab[k]))
161     longest_token_bytes = vocab[longest_token_id]
162
163     print("\n 词表分析:")
164     print(f" 最长 Token 的 ID: {longest_token_id}")
165     print(f" 最长 Token 的字节长度: {len(longest_token_bytes)}")
166     try:
167         print(f" 最长 Token 的文本内容: {longest_token_bytes.decode('utf-8')!r}")
168     except UnicodeDecodeError:
169         print(f" 最长 Token (原始字节): {longest_token_bytes}")
170
171     print("\n最初的 5 次 Merge:")
172     for i in range(min(5, len(merges))):
173         p1, p2 = merges[i]

```

```
174     print(f" {p1} + {p2} -> {p1 + p2}")
```

A.3 Tokenizer

```

1  import pickle
2  import regex as re
3  from typing import Iterable, Iterator
4
5  # =====
6  # 1. 全局配置与正则定义
7  # =====
8  PATTERN = r"''('?:[sdmt]|ll|ve|re)| ?\p{L}+| ?\p{N}+| ?[^\s\p{L}\p{N}]+\s+(?!\\S)\\s+''"
9  PRETOKEN_REGEX = re.compile(PATTERN)
10
11 # =====
12 # 2. Tokenizer 核心类
13 # =====
14 class Tokenizer:
15     def __init__(self, vocab: dict[int, bytes], merges: list[tuple[bytes, bytes]], special_tokens: list[str] |
        ↳ None = None):
16         """
17         初始化 Tokenizer, 建立查找索引并处理特殊 Token。
18         """
19         self.vocab = vocab.copy()
20         self.merges = merges
21         self.special_tokens = special_tokens or []
22
23         # 建立反向查找字典, 用于快速获取 token 对应的 ID
24         self.token_to_id = {v: k for k, v in self.vocab.items()}
25
26         # 将 merges 列表转化为 {pair: rank} 字典, 以 O(1) 的时间复杂度获取合并优先级
27         self.merge_ranks = {pair: i for i, pair in enumerate(self.merges)}
28
29         # 动态处理传入的 special_tokens: 如果不在词表中, 则追加到末尾
30         next_id = max(self.vocab.keys()) + 1 if self.vocab else 0
31         for st in self.special_tokens:
32             st_bytes = st.encode("utf-8")
33             if st_bytes not in self.token_to_id:
34                 self.vocab[next_id] = st_bytes
35                 self.token_to_id[st_bytes] = next_id
36                 next_id += 1
37
38         # 编译用于切分特殊 Token 的正则表达式
39         self.st_pattern = None
40         if self.special_tokens:
41             # 使用捕获组 (...), 这样 re.split 时能保留特殊 token 在列表中
42             sorted_st = sorted(self.special_tokens, key=len, reverse=True)
43             pattern_str = "|".join(map(re.escape, sorted_st))
44             self.st_pattern = re.compile(f"({pattern_str})")
45
46     @classmethod
47     def from_files(cls, vocab_filepath: str, merges_filepath: str, special_tokens: list[str] | None = None):
48         """
49         从序列化的 pickle 文件中加载 vocab 和 merges 并实例化 Tokenizer。
50         """
51         with open(vocab_filepath, "rb") as f:
52             vocab = pickle.load(f)
53         with open(merges_filepath, "rb") as f:
54             merges = pickle.load(f)
55
56         return cls(vocab, merges, special_tokens)

```

```

57
58 def _bpe_encode_chunk(self, chunk_text: str) -> list[int]:
59     """
60     内部辅助函数：对单个正则切割后的纯文本 pre-token 进行 BPE 合并操作。
61     """
62     # 将字符串打散成单字节列表
63     b_list = [bytes([b]) for b in chunk_text.encode("utf-8")]
64
65     # 如果长度为 1，直接返回（基础 256 字节必然在词表中）
66     if len(b_list) <= 1:
67         return [self.token_to_id[b] for b in b_list]
68
69     while True:
70         # 找到当前序列中，在 merges 中优先级最高（rank 最小）的相邻字节对
71         min_rank = float('inf')
72         best_pair = None
73
74         for i in range(len(b_list) - 1):
75             pair = (b_list[i], b_list[i+1])
76             rank = self.merge_ranks.get(pair)
77             if rank is not None and rank < min_rank:
78                 min_rank = rank
79                 best_pair = pair
80
81         # 如果没有任何一对在 merge_ranks 中，说明合并彻底完成
82         if best_pair is None:
83             break
84
85         # 执行最高优先级的一轮合并
86         new_b_list = []
87         i = 0
88         while i < len(b_list):
89             # 命中目标 pair
90             if i < len(b_list) - 1 and b_list[i] == best_pair[0] and b_list[i+1] == best_pair[1]:
91                 new_b_list.append(best_pair[0] + best_pair[1])
92                 i += 2
93             else:
94                 new_b_list.append(b_list[i])
95                 i += 1
96
97         b_list = new_b_list
98
99         # 如果只剩下一个完整的 token，提前退出
100         if len(b_list) == 1:
101             break
102
103         # 拿着最终合并好的字节序列，去词表里换取对应的 IDs
104         return [self.token_to_id[b] for b in b_list]
105
106 def encode(self, text: str) -> list[int]:
107     """
108     将整段输入文本编码为 token IDs。
109     """
110     ids = []
111
112     # 1. 全局截断：优先将文本通过 special_tokens 切开
113     if self.st_pattern:
114         parts = self.st_pattern.split(text)
115     else:

```

```

116         parts = [text]
117
118     for part in parts:
119         if not part:
120             continue
121
122         # 2. 如果这块 part 是一个 special token, 直接查 ID
123         if part in self.special_tokens:
124             ids.append(self.token_to_id[part.encode("utf-8")])
125         else:
126             # 3. 对普通文本进行预分词匹配
127             for match in PRETOKEN_REGEX.finditer(part):
128                 chunk_text = match.group()
129                 ids.extend(self._bpe_encode_chunk(chunk_text))
130
131     return ids
132
133 def encode_iterable(self, iterable: Iterable[str]) -> Iterator[int]:
134     """
135     懒加载生成器：逐块/逐行读取流数据并输出 IDs, 确保内存开销维持在常数级别。
136     说明：对于标准的文件句柄 (File Handle), 每次 yield 的是一行。
137     换行符属于 BPE 空格家族, 因此按行拆分不会截断合法的词组边界。
138     """
139     for chunk in iterable:
140         yield from self.encode(chunk)
141
142 def decode(self, ids: list[int]) -> str:
143     """
144     将整数 ID 序列还原为字符串。非法字节将被自动替换为 U+FFFD ()。
145     """
146     raw_bytes = b"".join(self.vocab[idx] for idx in ids)
147     return raw_bytes.decode("utf-8", errors="replace")

```

A.4 Adapters

```

1  from __future__ import annotations
2
3  import os
4  import math
5  import collections
6  import json
7  import regex
8  from collections.abc import Iterable
9  from typing import IO, Any, BinaryIO, Dict, List, Tuple
10
11 import numpy.typing as npt
12 import torch
13 import torch.nn.functional as F
14 import einx
15 from jaxtyping import Bool, Float, Int
16 from torch import Tensor
17 from collections import defaultdict
18 import pickle
19
20 from cs336_basics.test_function.bpe_trainer import train_bpe
21 from cs336_basics.test_function.tokenizer import Tokenizer
22 from cs336_basics.Chap_2.linear import Linear
23 from cs336_basics.Chap_2.embedding import Embedding
24 from cs336_basics.Chap_2.rmsnorm import RMSNorm
25 from cs336_basics.Chap_2.swiglu import SwiGLU
26 from cs336_basics.Chap_2.rope import RotaryPositionalEmbedding
27 from cs336_basics.Chap_2.attention import MultiHeadAttention
28
29
30 def run_linear(
31     d_in: int,
32     d_out: int,
33     weights: Float[Tensor, " d_out d_in"],
34     in_features: Float[Tensor, " ... d_in"],
35 ) -> Float[Tensor, " ... d_out"]:
36     """
37     Given the weights of a Linear layer, compute the transformation of a batched input.
38
39     Args:
40         in_dim (int): The size of the input dimension
41         out_dim (int): The size of the output dimension
42         weights (Float[Tensor, "d_out d_in"]): The linear weights to use
43         in_features (Float[Tensor, "... d_in"]): The output tensor to apply the function to
44
45     Returns:
46         Float[Tensor, "... d_out"]: The transformed output of your linear module.
47     """
48
49     model = Linear(
50         in_features=d_in,
51         out_features=d_out,
52         device=in_features.device,
53         dtype=in_features.dtype,
54     )
55
56     model.load_state_dict({"W": weights}, strict=True)
57

```

```

58     model.eval()
59     with torch.no_grad():
60         output = model(in_features)
61
62     return output
63
64
65 def run_embedding(
66     vocab_size: int,
67     d_model: int,
68     weights: Float[Tensor, " vocab_size d_model"],
69     token_ids: Int[Tensor, " ..."],
70 ) -> Float[Tensor, " ... d_model"]:
71     """
72     Given the weights of an Embedding layer, get the embeddings for a batch of token ids.
73
74     Args:
75         vocab_size (int): The number of embeddings in the vocabulary
76         d_model (int): The size of the embedding dimension
77         weights (Float[Tensor, "vocab_size d_model"]): The embedding vectors to fetch from
78         token_ids (Int[Tensor, "..."]): The set of token ids to fetch from the Embedding layer
79
80     Returns:
81         Float[Tensor, "... d_model"]: Batch of embeddings returned by your Embedding layer.
82     """
83
84     model = Embedding(
85         num_embeddings=vocab_size,
86         embedding_dim=d_model,
87         device=weights.device,
88         dtype=weights.dtype
89     )
90
91     model.load_state_dict({"weight": weights}, strict=True)
92
93     model.eval()
94     with torch.no_grad():
95         output = model(token_ids)
96
97     return output
98
99
100 def run_swiglu(
101     d_model: int,
102     d_ff: int,
103     w1_weight: Float[Tensor, " d_ff d_model"],
104     w2_weight: Float[Tensor, " d_model d_ff"],
105     w3_weight: Float[Tensor, " d_ff d_model"],
106     in_features: Float[Tensor, " ... d_model"],
107 ) -> Float[Tensor, " ... d_model"]:
108     """Given the weights of a SwiGLU network, return
109     the output of your implementation with these weights.
110
111     Args:
112         d_model (int): Dimensionality of the feedforward input and output.
113         d_ff (int): Dimensionality of the up-project happening internally to your swiglu.
114         w1_weight (Float[Tensor, "d_ff d_model"]): Stored weights for W1
115         w2_weight (Float[Tensor, "d_model d_ff"]): Stored weights for W2
116         w3_weight (Float[Tensor, "d_ff d_model"]): Stored weights for W3

```

```

117         in_features (Float[Tensor, "... d_model"]): Input embeddings to the feed-forward layer.
118
119     Returns:
120         Float[Tensor, "... d_model"]: Output embeddings of the same shape as the input embeddings.
121     """
122     # Example:
123     # If your state dict keys match, you can use `load_state_dict()`
124     # swiglu.load_state_dict(weights)
125     # You can also manually assign the weights
126     # swiglu.w1.weight.data = w1_weight
127     # swiglu.w2.weight.data = w2_weight
128     # swiglu.w3.weight.data = w3_weight
129     model = SwiGLU(
130         d_model=d_model,
131         d_ff=d_ff,
132         device=in_features.device,
133         dtype=in_features.dtype
134     )
135
136     state_dict = {
137         "w1.W": w1_weight,
138         "w2.W": w2_weight,
139         "w3.W": w3_weight
140     }
141     model.load_state_dict(state_dict, strict=True)
142
143     model.eval()
144     with torch.no_grad():
145         output = model(in_features)
146
147     return output
148
149
150 def run_scaled_dot_product_attention(
151     Q: Float[Tensor, "... queries d_k"],
152     K: Float[Tensor, "... keys d_k"],
153     V: Float[Tensor, "... values d_v"],
154     mask: Bool[Tensor, "... queries keys"] | None = None,
155 ) -> Float[Tensor, "... queries d_v"]:
156     """
157     Given key (K), query (Q), and value (V) tensors, return
158     the output of your scaled dot product attention implementation.
159
160     Args:
161         Q (Float[Tensor, "... queries d_k"]): Query tensor
162         K (Float[Tensor, "... keys d_k"]): Key tensor
163         V (Float[Tensor, "... values d_v"]): Values tensor
164         mask (Bool[Tensor, "... queries keys"] | None): Mask tensor
165
166     Returns:
167         Float[Tensor, "... queries d_v"]: Output of SDPA
168     """
169     # 提取 d_k 的维度大小, 用于后续的缩放
170     d_k = Q.size(-1)
171
172     # 1. 计算打分矩阵并缩放: (Q @ K^T) / sqrt(d_k)
173     # K.transpose(-2, -1) 的魔法: 只反转最后两个维度 (keys 和 d_k),
174     # 完全无视前面有多少个 batch 维度 (...), 极其优雅安全!
175     # 结果 scores 的形状将会是: (... queries, keys)
176     scores = (Q @ K.transpose(-2, -1)) / math.sqrt(d_k)

```

```

176
177 # 2. 应用掩码 (Masking)
178 if mask is not None:
179     # mask 中值为 True 表示“允许看”，False 表示“遮挡不允许看”。
180     # ~mask 取反后，把所有不允许看的位置强制填充为 -inf
181     scores = scores.masked_fill(~mask, float('-inf'))
182
183 # 3. Softmax 归一化
184 # 沿着最后一个维度 (keys) 将得分转化为概率分布。
185 # 这里你可以调用上一题自己手写的 run_softmax(scores, dim=-1),
186 # 但为了保证在多维 Tensor 上的极致性能，直接调用内置的 torch.softmax 也是绝佳选择。
187 attention_weights = torch.softmax(scores, dim=-1)
188
189 # 4. 乘以 V (提取信息)
190 # attention_weights 形状: (... queries, keys)
191 # V 形状: (... keys, d_v)
192 # 矩阵乘法后得到最终结果形状: (... queries, d_v)
193 output = attention_weights @ V
194
195 return output
196
197
198 def run_multihead_self_attention(
199     d_model: int,
200     num_heads: int,
201     q_proj_weight: Float[Tensor, "d_k d_in"],
202     k_proj_weight: Float[Tensor, "d_k d_in"],
203     v_proj_weight: Float[Tensor, "d_v d_in"],
204     o_proj_weight: Float[Tensor, "d_model d_v"],
205     in_features: Float[Tensor, "... sequence_length d_in"],
206 ) -> Float[Tensor, "... sequence_length d_out"]:
207     """
208     Given the key, query, and value projection weights of a naive unbatched
209     implementation of multi-head attention, return the output of an optimized batched
210     implementation. This implementation should handle the key, query, and value projections
211     for all heads in a single matrix multiply.
212     This function should not use RoPE.
213     See section 3.2.2 of Vaswani et al., 2017.
214
215     Args:
216         d_model (int): Dimensionality of the feedforward input and output.
217         num_heads (int): Number of heads to use in multi-headed attention.
218         max_seq_len (int): Maximum sequence length to pre-cache if your implementation does that.
219         q_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the Q projection
220         k_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the K projection
221         v_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the V projection
222         o_proj_weight (Float[Tensor, "d_model d_v"]): Weights for the output projection
223         in_features (Float[Tensor, "... sequence_length d_in"]): Tensor to run your implementation on.
224
225     Returns:
226         Float[Tensor, "... sequence_length d_out"]: Tensor with the output of running your optimized, batched
227         ↪ multi-headed attention
228         implementation with the given QKV projection weights and input features.
229     """
230 # 1. 实例化
231 model = MultiHeadAttention(
232     d_model=d_model,
233     num_heads=num_heads,
234     device=in_features.device,

```

```

234         dtype=in_features.dtype
235     )
236
237     # 2. 加载权重 (由于我们的 Linear 类内部参数叫 W, 所以加 .W 后缀)
238     state_dict = {
239         "q_proj.W": q_proj_weight,
240         "k_proj.W": k_proj_weight,
241         "v_proj.W": v_proj_weight,
242         "o_proj.W": o_proj_weight
243     }
244     model.load_state_dict(state_dict, strict=True)
245
246     # 3. 执行推理
247     model.eval()
248     with torch.no_grad():
249         output = model(in_features)
250
251     return output
252
253
254 def run_multihead_self_attention_with_rope(
255     d_model: int,
256     num_heads: int,
257     max_seq_len: int,
258     theta: float,
259     q_proj_weight: Float[Tensor, " d_k d_in"],
260     k_proj_weight: Float[Tensor, " d_k d_in"],
261     v_proj_weight: Float[Tensor, " d_v d_in"],
262     o_proj_weight: Float[Tensor, " d_model d_v"],
263     in_features: Float[Tensor, " ... sequence_length d_in"],
264     token_positions: Int[Tensor, " ... sequence_length"] | None = None,
265 ) -> Float[Tensor, " ... sequence_length d_out"]:
266     """
267     Given the key, query, and value projection weights of a naive unbatched
268     implementation of multi-head attention, return the output of an optimized batched
269     implementation. This implementation should handle the key, query, and value projections
270     for all heads in a single matrix multiply.
271     This version of MHA should include RoPE.
272     In this case, the RoPE embedding dimension must be the head embedding dimension (d_model // num_heads).
273     See section 3.2.2 of Vaswani et al., 2017.
274
275     Args:
276         d_model (int): Dimensionality of the feedforward input and output.
277         num_heads (int): Number of heads to use in multi-headed attention.
278         max_seq_len (int): Maximum sequence length to pre-cache if your implementation does that.
279         theta (float): RoPE parameter.
280         q_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the Q projection
281         k_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the K projection
282         v_proj_weight (Float[Tensor, "d_k d_in"]): Weights for the V projection
283         o_proj_weight (Float[Tensor, "d_model d_v"]): Weights for the output projection
284         in_features (Float[Tensor, "... sequence_length d_in"]): Tensor to run your implementation on.
285         token_positions (Int[Tensor, " ... sequence_length"] | None): Optional tensor with the positions of the
286             ↪ tokens
287
288     Returns:
289         Float[Tensor, " ... sequence_length d_out"]: Tensor with the output of running your optimized, batched
290             ↪ multi-headed attention
291         implementation with the given QKV projection weights and input features.
292     """

```

```

291     # 1. 维度准备
292     d_head = d_model // num_heads
293     seq_len = in_features.shape[-2]
294
295     # 2. 线性投影与多头切分 (一步位移)
296     # 直接投影后切成 [... heads, seq, d_head]
297     Q = einx.rearrange("... s (h d) -> ... h s d", F.linear(in_features, q_proj_weight), h=num_heads)
298     K = einx.rearrange("... s (h d) -> ... h s d", F.linear(in_features, k_proj_weight), h=num_heads)
299     V = einx.rearrange("... s (h d) -> ... h s d", F.linear(in_features, v_proj_weight), h=num_heads)
300
301     # 3. 注入 RoPE
302     rope = RotaryPositionalEmbedding(theta, d_head, max_seq_len, device=in_features.device)
303     if token_positions is None:
304         token_positions = torch.arange(seq_len, device=in_features.device)
305
306     # 利用 unsqueeze(-2) 配合 RoPE 内部的 einx 逻辑进行 heads 维度的广播
307     pos_for_rope = token_positions.unsqueeze(-2)
308     Q, K = rope(Q, pos_for_rope), rope(K, pos_for_rope)
309
310     # 4. 【核心优化】计算注意力打分 (einx.dot)
311     # 代替 (Q @ K.transpose(-2, -1))
312     # 语义: Query(s_q) 与 Key(s_k) 在特征维 (d) 点积, 保留 batch(...) 和 head(h)
313     logits = einx.dot("... h s_q d, ... h s_k d -> ... h s_q s_k", Q, K) / math.sqrt(d_head)
314
315     # 5. 因果掩码与归一化
316     # 这里的 mask 也可以用 einx 思想理解: 它只作用于最后两个 s 维度
317     mask = torch.tril(torch.ones(seq_len, seq_len, dtype=torch.bool, device=in_features.device))
318     # 使用 torch.where 配合布尔掩码比 masked_fill 更符合函数式风格
319     attn_weights = torch.softmax(torch.where(mask, logits, float("-inf")), dim=-1)
320
321     # 6. 【核心优化】加权聚合提取 Value (einx.dot)
322     # 代替 attn_weights @ V
323     # 语义: 用权重 (s_k) 对内容 (s_k) 进行加权求和, 产出查询位置 (s_q) 的特征 (d)
324     out = einx.dot("... h s_q s_k, ... h s_k d -> ... h s_q d", attn_weights, V)
325
326     # 7. 缝合多头并最终投影
327     out = einx.rearrange("... h s d -> ... s (h d)", out)
328     return F.linear(out, o_proj_weight)
329
330
331 def run_rope(
332     d_k: int,
333     theta: float,
334     max_seq_len: int,
335     in_query_or_key: Float[Tensor, "... sequence_length d_k"],
336     token_positions: Int[Tensor, "... sequence_length"],
337 ) -> Float[Tensor, "... sequence_length d_k"]:
338     """
339     Run RoPE for a given input tensor.
340
341     Args:
342         d_k (int): Embedding dimension size for the query or key tensor.
343         theta (float): RoPE parameter.
344         max_seq_len (int): Maximum sequence length to pre-cache if your implementation does that.
345         in_query_or_key (Float[Tensor, "... sequence_length d_k"]): Input tensor to run RoPE on.
346         token_positions (Int[Tensor, "... sequence_length"]): Tensor of shape (batch_size, sequence_length) with
347             ↪ the token positions
348
349     Returns:
350         Float[Tensor, "... sequence_length d_k"]: Tensor with RoPEd input.

```

```

349     """
350     # 1. 实例化 RoPE 模块
351     model = RotaryPositionalEmbedding(
352         theta=theta,
353         d_k=d_k,
354         max_seq_len=max_seq_len,
355         device=in_query_or_key.device
356     )
357
358
359     # 2. 评估模式并切断梯度，直接计算
360     model.eval()
361     with torch.no_grad():
362         # 注意要把 token_positions 传进去！
363         output = model(in_query_or_key, token_positions)
364
365     return output
366
367
368 def run_transformer_block(
369     d_model: int,
370     num_heads: int,
371     d_ff: int,
372     max_seq_len: int,
373     theta: float,
374     weights: dict[str, Tensor],
375     in_features: Float[Tensor, " batch sequence_length d_model"],
376 ) -> Float[Tensor, " batch sequence_length d_model"]:
377     """
378     Given the weights of a pre-norm Transformer block and input features,
379     return the output of running the Transformer block on the input features.
380
381     This function should use RoPE.
382     Depending on your implementation, you may simply need to pass the relevant args
383     to your TransformerBlock constructor, or you may need to initialize your own RoPE
384     class and pass that instead.
385
386     Args:
387         d_model (int): The dimensionality of the Transformer block input.
388         num_heads (int): Number of heads to use in multi-headed attention. `d_model` must be
389             evenly divisible by `num_heads`.
390         d_ff (int): Dimensionality of the feed-forward inner layer.
391         max_seq_len (int): Maximum sequence length to pre-cache if your implementation does that.
392         theta (float): RoPE parameter.
393         weights (dict[str, Tensor]):
394             State dict of our reference implementation.
395             The keys of this dictionary are:
396             - `attn.q_proj.weight`
397                 The query projections for all `num_heads` attention heads.
398                 Shape is (d_model, d_model).
399                 The rows are ordered by matrices of shape (num_heads, d_k),
400                 so `attn.q_proj.weight == torch.cat([q_heads.0.weight, ..., q_heads.N.weight], dim=0)`.
401             - `attn.k_proj.weight`
402                 The key projections for all `num_heads` attention heads.
403                 Shape is (d_model, d_model).
404                 The rows are ordered by matrices of shape (num_heads, d_k),
405                 so `attn.k_proj.weight == torch.cat([k_heads.0.weight, ..., k_heads.N.weight], dim=0)`.
406             - `attn.v_proj.weight`
407                 The value projections for all `num_heads` attention heads.

```

```

408         Shape is (d_model, d_model).
409         The rows are ordered by matrices of shape (num_heads, d_v),
410         so `attn.v_proj.weight == torch.cat([v_heads.0.weight, ..., v_heads.N.weight], dim=0)`.
411     - `attn.output_proj.weight`
412         Weight of the multi-head self-attention output projection
413         Shape is (d_model, d_model).
414     - `ln1.weight`
415         Weights of affine transform for the first RMSNorm
416         applied in the transformer block.
417         Shape is (d_model,).
418     - `ffn.w1.weight`
419         Weight of the first linear transformation in the FFN.
420         Shape is (d_model, d_ff).
421     - `ffn.w2.weight`
422         Weight of the second linear transformation in the FFN.
423         Shape is (d_ff, d_model).
424     - `ffn.w3.weight`
425         Weight of the third linear transformation in the FFN.
426         Shape is (d_model, d_ff).
427     - `ln2.weight`
428         Weights of affine transform for the second RMSNorm
429         applied in the transformer block.
430         Shape is (d_model,).
431     in_features (Float[Tensor, "batch sequence_length d_model"]):
432         Tensor to run your implementation on.
433
434     Returns:
435         Float[Tensor, "batch sequence_length d_model"] Tensor with the output of
436         running the Transformer block on the input features while using RoPE.
437     """
438     raise NotImplementedError
439
440
441 def run_transformer_lm(
442     vocab_size: int,
443     context_length: int,
444     d_model: int,
445     num_layers: int,
446     num_heads: int,
447     d_ff: int,
448     rope_theta: float,
449     weights: dict[str, Tensor],
450     in_indices: Int[Tensor, " batch_size sequence_length"],
451 ) -> Float[Tensor, " batch_size sequence_length vocab_size"]:
452     r"""Given the weights of a Transformer language model and input indices,
453     return the output of running a forward pass on the input indices.
454
455     This function should use RoPE.
456
457     Args:
458         vocab_size (int): The number of unique items in the output vocabulary to be predicted.
459         context_length (int): The maximum number of tokens to process at once.
460         d_model (int): The dimensionality of the model embeddings and sublayer outputs.
461         num_layers (int): The number of Transformer layers to use.
462         num_heads (int): Number of heads to use in multi-headed attention. `d_model` must be
463             evenly divisible by `num_heads`.
464         d_ff (int): Dimensionality of the feed-forward inner layer (section 3.3).
465         rope_theta (float): The RoPE  $\Theta$  parameter.
466         weights (dict[str, Tensor]):

```

```

467     State dict of our reference implementation. {num_layers} refers to an
468     integer between `0` and `num_layers - 1` (the layer index).
469     The keys of this dictionary are:
470     - `token_embeddings.weight`
471         Token embedding matrix. Shape is (vocab_size, d_model).
472     - `layers.{num_layers}.attn.q_proj.weight`
473         The query projections for all `num_heads` attention heads.
474         Shape is (num_heads * (d_model / num_heads), d_model).
475         The rows are ordered by matrices of shape (num_heads, d_k),
476         so `attn.q_proj.weight == torch.cat([q_heads.0.weight, ..., q_heads.N.weight], dim=0)`.
477     - `layers.{num_layers}.attn.k_proj.weight`
478         The key projections for all `num_heads` attention heads.
479         Shape is (num_heads * (d_model / num_heads), d_model).
480         The rows are ordered by matrices of shape (num_heads, d_k),
481         so `attn.k_proj.weight == torch.cat([k_heads.0.weight, ..., k_heads.N.weight], dim=0)`.
482     - `layers.{num_layers}.attn.v_proj.weight`
483         The value projections for all `num_heads` attention heads.
484         Shape is (num_heads * (d_model / num_heads), d_model).
485         The rows are ordered by matrices of shape (num_heads, d_v),
486         so `attn.v_proj.weight == torch.cat([v_heads.0.weight, ..., v_heads.N.weight], dim=0)`.
487     - `layers.{num_layers}.attn.output_proj.weight`
488         Weight of the multi-head self-attention output projection
489         Shape is ((d_model / num_heads) * num_heads, d_model).
490     - `layers.{num_layers}.ln1.weight`
491         Weights of affine transform for the first RMSNorm
492         applied in the transformer block.
493         Shape is (d_model,).
494     - `layers.{num_layers}.ffn.w1.weight`
495         Weight of the first linear transformation in the FFN.
496         Shape is (d_model, d_ff).
497     - `layers.{num_layers}.ffn.w2.weight`
498         Weight of the second linear transformation in the FFN.
499         Shape is (d_ff, d_model).
500     - `layers.{num_layers}.ffn.w3.weight`
501         Weight of the third linear transformation in the FFN.
502         Shape is (d_model, d_ff).
503     - `layers.{num_layers}.ln2.weight`
504         Weights of affine transform for the second RMSNorm
505         applied in the transformer block.
506         Shape is (d_model,).
507     - `ln_final.weight`
508         Weights of affine transform for RMSNorm applied to the output of the final transformer block.
509         Shape is (d_model, ).
510     - `lm_head.weight`
511         Weights of the language model output embedding.
512         Shape is (vocab_size, d_model).
513     in_indices (Int[Tensor, "batch_size sequence_length"]) Tensor with input indices to run the language model
514     ↪ on. Shape is (batch_size, sequence_length), where
515         `sequence_length` is at most `context_length`.
516
517 Returns:
518     Float[Tensor, "batch_size sequence_length vocab_size"]: Tensor with the predicted unnormalized
519     next-word distribution for each token.
520 """
521 raise NotImplementedError
522
523 def run_rmsnorm(
524     d_model: int,

```

```

525     eps: float,
526     weights: Float[Tensor, " d_model"],
527     in_features: Float[Tensor, " ... d_model"],
528 ) -> Float[Tensor, " ... d_model"]:
529     """Given the weights of a RMSNorm affine transform,
530     return the output of running RMSNorm on the input features.
531
532     Args:
533         d_model (int): The dimensionality of the RMSNorm input.
534         eps: (float): A value added to the denominator for numerical stability.
535         weights (Float[Tensor, "d_model"]): RMSNorm weights.
536         in_features (Float[Tensor, "... d_model"]): Input features to run RMSNorm on. Can have arbitrary leading
537             dimensions.
538
539     Returns:
540         Float[Tensor, "... d_model"]: Tensor of with the same shape as `in_features` with the output of running
541         RMSNorm of the `in_features`.
542     """
543     model = RMSNorm(
544         d_model=d_model,
545         eps=eps,
546         device=weights.device,
547         dtype=weights.dtype
548     )
549
550     model.load_state_dict({"weight": weights}, strict=True)
551
552     model.eval()
553     with torch.no_grad():
554         output = model(in_features)
555
556     return output
557
558
559 def run_silu(in_features: Float[Tensor, " ..."]) -> Float[Tensor, " ..."]:
560     """Given a tensor of inputs, return the output of applying SiLU
561     to each element.
562
563     Args:
564         in_features (Float[Tensor, "..."]): Input features to run SiLU on. Shape is arbitrary.
565
566     Returns:
567         Float[Tensor, "..."]: of with the same shape as `in_features` with the output of applying
568         SiLU to each element.
569     """
570     raise NotImplementedError
571
572
573 def run_get_batch(
574     dataset: npt.NDArray, batch_size: int, context_length: int, device: str
575 ) -> tuple[torch.Tensor, torch.Tensor]:
576     """
577     Given a dataset (a 1D numpy array of integers) and a desired batch size and
578     context length, sample language modeling input sequences and their corresponding
579     labels from the dataset.
580
581     Args:
582         dataset (np.array): 1D numpy array of integer token IDs in the dataset.
583         batch_size (int): Desired batch size to sample.

```

```

584         context_length (int): Desired context length of each sampled example.
585         device (str): PyTorch device string (e.g., 'cpu' or 'cuda:0') indicating the device
586             to place the sampled input sequences and labels on.
587
588     Returns:
589         Tuple of torch.LongTensors of shape (batch_size, context_length). The first tuple item
590         is the sampled input sequences, and the second tuple item is the corresponding
591         language modeling labels.
592     """
593     raise NotImplementedError
594
595
596 def run_softmax(in_features: Float[Tensor, "..."], dim: int) -> Float[Tensor, "..."]:
597     """
598     Given a tensor of inputs, return the output of softmaxing the given `dim`
599     of the input.
600
601     Args:
602         in_features (Float[Tensor, "..."]): Input features to softmax. Shape is arbitrary.
603         dim (int): Dimension of the `in_features` to apply softmax to.
604
605     Returns:
606         Float[Tensor, "..."]: Tensor of with the same shape as `in_features` with the output of
607         softmax normalizing the specified `dim`.
608     """
609     # 1. 寻找指定维度上的最大值
610     # keepdim=True 是灵魂! 它能保留维度形状 (比如把形状从 [3, 4] 变成 [3, 1]),
611     # 这样后续减法时才能正确触发广播机制 (Broadcasting)。
612     # 注意: torch.max 会返回一个元组 (values, indices), 我们只需要取第 0 项的 values。
613     max_vals = in_features.max(dim=dim, keepdim=True)[0]
614
615     # 2. Max Trick: 平移输入值, 防止指数爆炸
616     shifted_logits = in_features - max_vals
617
618     # 3. 计算指数 (现在所有的输入都 <= 0, 所以 exp 的结果都在 0~1 之间, 绝对安全)
619     exps = torch.exp(shifted_logits)
620
621     # 4. 沿着指定维度求和, 作为分母
622     sum_exps = exps.sum(dim=dim, keepdim=True)
623
624     # 5. 归一化, 得到最终的概率分布
625     probabilities = exps / sum_exps
626
627     return probabilities
628
629
630 def run_cross_entropy(
631     inputs: Float[Tensor, "batch_size vocab_size"], targets: Int[Tensor, "batch_size"]
632 ) -> Float[Tensor, ""]:
633     """Given a tensor of inputs and targets, compute the average cross-entropy
634     loss across examples.
635
636     Args:
637         inputs (Float[Tensor, "batch_size vocab_size"]): inputs[i][j] is the
638             unnormalized logit of jth class for the ith example.
639         targets (Int[Tensor, "batch_size"]): Tensor of shape (batch_size,) with the index of the correct class.
640             Each value must be between 0 and `num_classes - 1`.
641
642     Returns:

```

```

643         Float[Tensor, ""]: The average cross-entropy loss across examples.
644     """
645     raise NotImplementedError
646
647
648 def run_gradient_clipping(parameters: Iterable[torch.nn.Parameter], max_l2_norm: float) -> None:
649     """Given a set of parameters, clip their combined gradients to have l2 norm at most max_l2_norm.
650
651     Args:
652         parameters (Iterable[torch.nn.Parameter]): collection of trainable parameters.
653         max_l2_norm (float): a positive value containing the maximum l2-norm.
654
655     The gradients of the parameters (parameter.grad) should be modified in-place.
656     """
657     raise NotImplementedError
658
659
660 def get_adamw_cls() -> Any:
661     """
662     Returns a torch.optim.Optimizer that implements AdamW.
663     """
664     raise NotImplementedError
665
666
667 def run_get_lr_cosine_schedule(
668     it: int,
669     max_learning_rate: float,
670     min_learning_rate: float,
671     warmup_iters: int,
672     cosine_cycle_iters: int,
673 ):
674     """
675     Given the parameters of a cosine learning rate decay schedule (with linear
676     warmup) and an iteration number, return the learning rate at the given
677     iteration under the specified schedule.
678
679     Args:
680         it (int): Iteration number to get learning rate for.
681         max_learning_rate (float): alpha_max, the maximum learning rate for
682             cosine learning rate schedule (with warmup).
683         min_learning_rate (float): alpha_min, the minimum / final learning rate for
684             the cosine learning rate schedule (with warmup).
685         warmup_iters (int): T_w, the number of iterations to linearly warm-up
686             the learning rate.
687         cosine_cycle_iters (int): T_c, the number of cosine annealing iterations.
688
689     Returns:
690         Learning rate at the given iteration under the specified schedule.
691     """
692     raise NotImplementedError
693
694
695 def run_save_checkpoint(
696     model: torch.nn.Module,
697     optimizer: torch.optim.Optimizer,
698     iteration: int,
699     out: str | os.PathLike | BinaryIO | IO[bytes],
700 ):
701     """

```

```

702     Given a model, optimizer, and an iteration number, serialize them to disk.
703
704     Args:
705         model (torch.nn.Module): Serialize the state of this model.
706         optimizer (torch.optim.Optimizer): Serialize the state of this optimizer.
707         iteration (int): Serialize this value, which represents the number of training iterations
708             we've completed.
709         out (str | os.PathLike | BinaryIO | IO[bytes]): Path or file-like object to serialize the model,
710             ↪ optimizer, and iteration to.
711
712     """
713     raise NotImplementedError
714
715 def run_load_checkpoint(
716     src: str | os.PathLike | BinaryIO | IO[bytes],
717     model: torch.nn.Module,
718     optimizer: torch.optim.Optimizer,
719 ) -> int:
720     """
721     Given a serialized checkpoint (path or file-like object), restore the
722     serialized state to the given model and optimizer.
723     Return the number of iterations that we previously serialized in
724     the checkpoint.
725
726     Args:
727         src (str | os.PathLike | BinaryIO | IO[bytes]): Path or file-like object to serialized checkpoint.
728         model (torch.nn.Module): Restore the state of this model.
729         optimizer (torch.optim.Optimizer): Restore the state of this optimizer.
730
731     Returns:
732         int: the previously-serialized number of iterations.
733
734     """
735     raise NotImplementedError
736
737 def get_tokenizer(
738     vocab: dict[int, bytes],
739     merges: list[tuple[bytes, bytes]],
740     special_tokens: list[str] | None = None,
741 ) -> Any:
742     """
743     Given a vocabulary, a list of merges, and a list of special tokens,
744     return a BPE tokenizer that uses the provided vocab, merges, and special tokens.
745
746     Args:
747         vocab (dict[int, bytes]): The tokenizer vocabulary, a mapping from int (token ID in the vocabulary)
748             to bytes (token bytes)
749         merges (list[tuple[bytes, bytes]]): BPE merges. Each list item is a tuple of bytes (<token1>, <token2>),
750             representing that <token1> was merged with <token2>.
751         Merges are ordered by order of creation.
752         special_tokens (list[str] | None): A list of string special tokens for the tokenizer. These strings will
753             ↪ never
754             be split into multiple tokens, and will always be kept as a single token.
755
756     Returns:
757         A BPE tokenizer that uses the provided vocab, merges, and special tokens.
758
759     """
760     return Tokenizer(vocab=vocab, merges=merges, special_tokens=special_tokens)
761
762 def run_train_bpe(

```

```
759     input_path: str,
760     vocab_size: int,
761     special_tokens: list[str]
762 ) -> tuple[dict[int, bytes], list[tuple[bytes, bytes]]]:
763     """
764     Adapter function for training the BPE tokenizer.
765     This routes the inputs from the test suite to our optimized implementation.
766     """
767
768     # 直接调用核心逻辑并返回结果
769     vocab, merges = train_bpe(
770         input_path=input_path,
771         vocab_size=vocab_size,
772         special_tokens=special_tokens
773     )
774
775     return vocab, merges
```
